

Unit: 2.

FUNCTION & LIMIT**SECTION: A (MULTIPLE CHOICE QUESTIONS)**

1. A function $f(x) = \frac{x}{|x|}, x \neq 0$ is a/an _____. (MP:26)
A) odd function. (b) even function (c) linear function (d) neither even nor odd
2. The interval $[4,5] \cap (4,5) =$: (2025)
 (a) $\emptyset$ (b) **(4, 5)** (c) $\{4, 5\}$ (d) $[4, 5]$
3. The interval $(2, \infty) \cap (1, 3) =$: (MP:25)
 (a) $[2, 3]$ (b) **(2, 3)** (c) $[2, 3)$ (d) $(2, 3]$
4. $[3,5] - (3,5) =$ (MP:24) (2024)
 (a) $\emptyset$ (b) $(3,5)$ (c) $[3,5]$ (d) **$\{3,5\}$**
5. The set $\{x|x \in R\}$ can be written as
 (a) $[-\infty, \infty]$ (b) $[-\infty, a]$ (c) **$(-\infty, \infty)$** (d) (a, ∞)
6. If $g(x) = 3x + 2$ and $g(f(x)) = x$ then $f(2) =$
 (a) 2 (b) 6 (c) **0** (d) 8
7. If $f(x) = 4^x$ then $f(x+1) - f(x) =$
 (a) 4 (b) $f(x)$ (c) **$3f(x)$** (d) $2f(x)$
8. $\lim_{x \rightarrow 3} \frac{x^2-9}{x-3} = ?$
 (a) 3 (b) ∞ (c) 0 (d) **6**
9. If " x " is in radian, $\frac{\sin ax}{x}$: (2025)
 (a) 0 (b) **a** (c) 1 (d) $\frac{1}{a}$
10. $\lim_{x \rightarrow 0} \frac{\sin x^0}{x}$: (MP:25)
 (a) 0 (b) 1 (c) $\frac{180}{\pi}$ (d) **$\frac{\pi}{180}$**
11. $\lim_{x \rightarrow \infty} (1+x)^{\frac{1}{x}}$: (MP:24) (MP:26)
 (a) e^3 (b) $e^{\frac{-1}{2}}$ (c) **e** (d) e^{-1}
12. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-7n} =$: (2024)
 (a) $-e$ (b) 1 (c) **e^{-7}** (d) ∞

SECTION: B (SHORT ANSWER QUESTIONS)**Exercise: 2.1**

1. Give that $f(x) = \frac{x+1}{3}$ and $g(x) = 3x + 5$ are two given bijective functions, then show that $(gof)^{-1} = f^{-1}og^{-1}$. (2025)
OR: Give that $f(x) = \frac{x+1}{2}$ and $g(x) = 2x - 1$ are two given bijective functions then show that $(gof)^{-1} = f^{-1}og^{-1}$. (MP:25)

Exercise: 2.3

2. Evaluate the limit: (Anyone)

(i) $\lim_{x \rightarrow 1} \frac{\frac{1}{\sqrt{x}} - 1}{1 - x}$ (MP:24)

(ii) $\lim_{x \rightarrow 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1}$

(iii) $\lim_{y \rightarrow 4} \frac{y^{\frac{5}{2}} - 16y^{\frac{1}{2}}}{y - 4}$

(iv) $\lim_{x \rightarrow 0} (1 + 2x)^{\frac{5}{x}}$

(v) $\lim_{x \rightarrow \infty} \left(1 + \frac{p}{x}\right)^{px}$

(vi) $\lim_{x \rightarrow 0} \left(1 - \frac{pq}{x}\right)^{\frac{x}{9}}$

(vii) $\lim_{x \rightarrow 0} \left(\frac{1+7x}{1-9x}\right)^{\frac{1}{x}}$

(viii) $\lim_{x \rightarrow 0} \frac{e^{-2x}e^{-11x}}{x}$ (MP:25)

(ix) $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$

(x) $\lim_{x \rightarrow 0} \frac{a^x + a^{-x} - 2}{x^2}$

(xi) $\lim_{x \rightarrow \infty} \frac{3e^{-5x} - 5e^{-2x} + 2}{x}$

(xii) $\lim_{x \rightarrow 0} \frac{2^x - 4^x - 8^x - 1}{x + 2}$

(xiii) $\lim_{x \rightarrow 0} \frac{6^x - 3^x - 2^x + 1}{x^2}$

(xiv) $\lim_{x \rightarrow 1} x^{\frac{1}{(x-1)}}$

(xv) $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{\sin^2 x}$ (2025)

(xvi) $\lim_{x \rightarrow 0} \frac{\sin x^0}{x}$ (2024)

(xvii) $\lim_{x \rightarrow 0} \frac{\sin 3x \sin 5x}{7x^2}$

(xviii) $\lim_{x \rightarrow 0} \frac{\sin^2\left(\frac{x}{2}\right)}{4x^2}$

(xix) $\lim_{x \rightarrow 0} \frac{1 - \cos mx}{1 - \cos nx}$

(xx) $\lim_{x \rightarrow 0} \frac{\sin px}{\sin qx}$

(xxi) $\lim_{x \rightarrow 0} \frac{(2\pi - x)\sec(\pi - x)}{\frac{\pi}{2}}$

(xxii) $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$

(xxiii) $\lim_{x \rightarrow \pi} \frac{\sqrt{5 + \cos x} - 2}{\pi - x}$ (MP:25)

(xxiv) $\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}$ (MP:24) (2024) (MP:25) (2025) (MP:26)

Ex: 2.4

3. Find the values of m and n , so that given function f is continuous at $x = 3$:

$$f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x + 9 & \text{if } x > 3 \end{cases} \quad \text{(MP:24)}$$

4. Test continuity and discontinuity of the following function:

$$f(x) = \begin{cases} 2 + x, & \text{when } x < 3 \\ 5 - 2x, & \text{when } x \geq 3 \end{cases} \quad \text{at } x = 3 \quad \text{(2024) (MP:26)}$$

5. Find the value of k if $f(x) = \begin{cases} kx - 9, & x < 5 \\ 9x - k, & x > 5 \\ 36, & x = 5 \end{cases}$ is continuous at $x = 5$.

6. If $f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{5+7}}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$, Find the value of k so that f is continuous at $x = 2$.

7. Test the continuity: $f(x) = \begin{cases} \frac{e^x - 1}{e^x + 1}, & \text{when } x \neq 0 \\ 1, & \text{when } x = 0 \end{cases}$

Farhan Yousuf Khan

M.E Education System